

Review: Linear just-identified IV

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- **Exclusion restriction:** $\text{Cov}(Z, u) = 0$ (instrument only affects Y through X)
- **Instrument exogeneity:** instrument is as-good-as-randomly assigned (often folded into the exclusion restriction)

Recent literature has raised the bar for what counts as “plausible”

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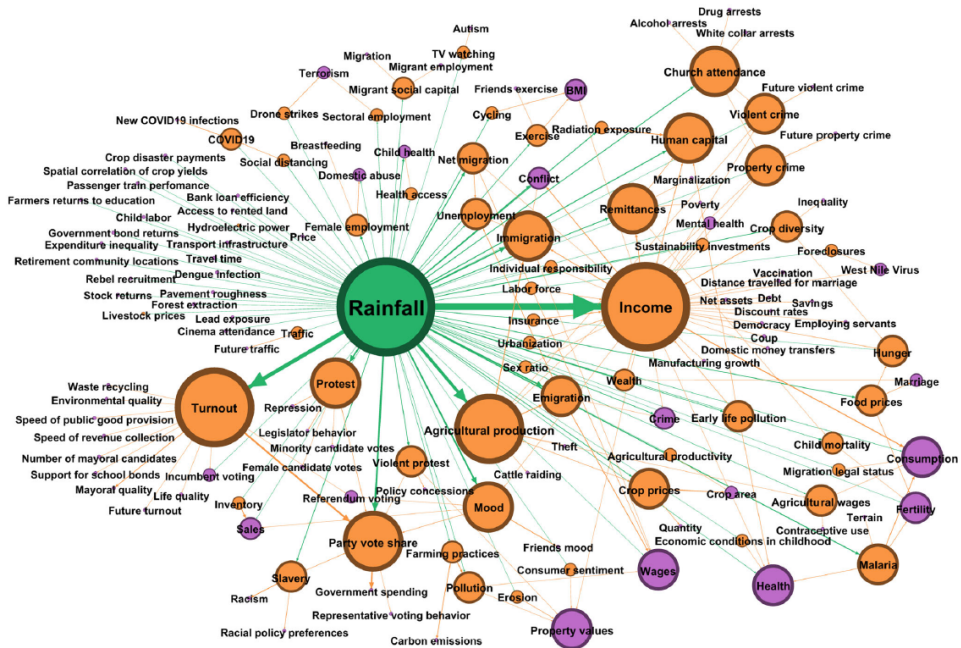
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2. Lee et al. (2022, *AER*): valid inference requires $F > 105$ rather than $F > 10$
3. Young (2022, *EER*): AEA papers show IV results highly sensitive to outliers

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- For weather IVs, scholars must assume hundreds of additional identifying assumptions
- Similar concerns likely apply to other commonly-used instruments

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- Introduce tF procedure: adjust critical values for 2nd-stage t based on 1st-stage F
- For 25% of AER specifications, corrected standard errors 49% larger at 5% level

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- Non-*iid* errors and high leverage worsen both size and power

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- First-stage F -statistics similarly sensitive to outliers
- Problems concentrated in high-leverage specifications

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- Walk away and admit that it's impossible to obtain a causal estimate
- Use [partial identification](#) methods to put bounds on causal effects:
 - “Upper” bound: biased OLS
 - “Lower” bound: plausible treatment effect